(2) Kirchhoff's Current Law (KCL) : The current flow into a node is equal to the current flow out of the node. The law can also be stated as the algebraic sum of the current entering any node is zero. Note that node is a point at which two or more elements of the circuit have a common connection.

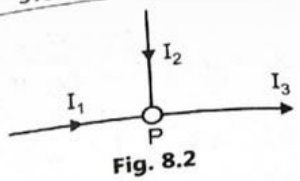


Fig. 8.2

By KCL, we get $I_1 + I_2 - I_3 = 0$ or $I_1 + I_2 = I_3$

(3) Kirchhoff's Voltage Law (KVL) : The algebraic sum of the voltage drops around a closed loop is equal to the to sources in the loop. Now, we consider some examples.

Illustrations **(Dec. 2019,**

Ex. 1 : Determine the currents in the network given in figure below.

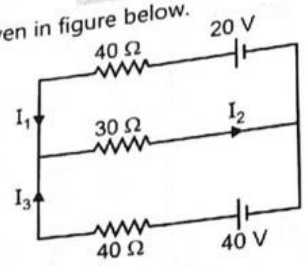


Fig. 8.3

Sol. : Applying KCL and KVL we get the equations

$I_1 + I_3 = I_2$ or $I_1 - I_2 + I_3 = 0$

$40I_1 + 30I_2 = 20$ $4I_1 + 3I_2 = 2$

$40I_3 + 30I_2 = 40$ $3I_2 + 4I_3 = 4$

Consider, augmented matrix

$$\begin{bmatrix} 1 & -1 & 1 & : & 0 \\ 4 & 3 & 0 & : & 2 \\ 0 & 3 & 4 & : & 4 \end{bmatrix} \quad R_2 - 4R_1 \sim \begin{bmatrix} 1 & -1 & 1 & : & 0 \\ 0 & 7 & -4 & : & 2 \\ 0 & 3 & 4 & : & 4 \end{bmatrix} \quad 7R_3 - 3R_2 \sim \begin{bmatrix} 1 & -1 & 1 & : & 0 \\ 0 & 7 & -4 & : & 2 \\ 0 & 0 & 40 & : & 22 \end{bmatrix}$$

Resulting equations are $I_1 - I_2 + I_3 = 0$ $I_2 = \frac{84}{140} = \frac{6}{10}$

$7I_2 - 4I_3 = 2$ $I_1 = \frac{1}{20}$

$40I_3 = 22$ $I_3 = \frac{22}{40} = \frac{11}{20}$

$\therefore \quad I_1 = \frac{1}{20}$, $I_2 = \frac{6}{10}$ and $I_3 = \frac{11}{20}$.

Ex. 2 : Find the currents in the circuit shown below.

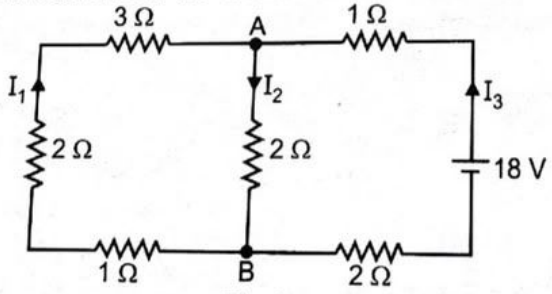


Fig. 8.4

Sol. : Applying KCL at nodes A and B, we get

$I_1 + I_3 = I_2$

$I_2 = I_1 + I_3$

Both are the equation, are same and we require additional two equations.

Then we get by applying KVL to left and right loops of the network.

For left loop $6I_1 + 2I_2 = 0$

For right loop $2I_2 + 3I_3 = 18$

Thus we get three equations

$$I_1 - I_2 + I_3 = 0$$
$$6I_1 + 2I_2 = 0$$
$$2I_2 + 3I_3 = 18$$

Writing in matrix for

$$\begin{bmatrix} 1 & -1 & 1 \\ 6 & 2 & 0 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 18 \end{bmatrix}$$

$$R_2 - 6R_1 \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 8 & -6 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 18 \end{bmatrix}$$

$$R_{23} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 3 \\ 0 & 8 & -6 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 18 \\ 0 \end{bmatrix}$$

$$R_3 - 4R_2 \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -18 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 18 \\ -72 \end{bmatrix}$$

Resulting equations are

$$I_1 - I_2 + I_3 = 0 \qquad I_1 = -1$$
$$2I_2 + 3I_3 = 18 \qquad 2I_2 + 12 = 18 \quad \therefore \quad I_2 = 3$$
$$-18I_3 = -72 \qquad I_3 = 4$$

$\therefore$ $I_1 = -1$ amp, $I_2 = 3$ amps and $I_3 = 4$ amps.

Direction of I_1 is opposite to that shown in the figure.

Ex. 3 : Find the currents in the various branches of the following network, shown in the Fig. 8.3.

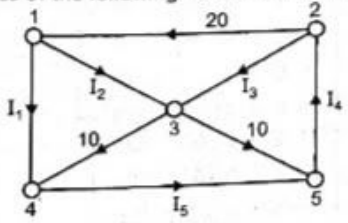


Fig. 8.3

Applying KCL for various nodes

At node 1	$I_1 + I_2 = 20$	$I_1 + I_2 = 20$
At node 2	$I_3 + 20 = I_4$	$I_3 - I_4 = -20$
At node 3	$I_2 + I_3 = 10 + 10$	$I_2 + I_3 = 20$
At node 4	$I_1 + 10 = I_5$	$I_1 - I_5 = -10$
At node 5	$I_5 + 10 = I_4$	$I_4 - I_5 = 10$

... (1)

Writing equation (1) in matrix form

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 20 \\ -20 \\ 20 \\ -10 \\ 10 \end{bmatrix}$$

$$AI = B$$

Consider, $[A : B] =$

$$\left[\begin{array}{ccccc:c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 1 & 1 & 0 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -10 \\ 0 & 0 & 0 & 1 & -1 & 10 \end{array}\right]$$

$$R_4 - R_1 \sim \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & : & 20 \\ 0 & 0 & 1 & -1 & 0 & : & -20 \\ 0 & 1 & 1 & 0 & 0 & : & 20 \\ 0 & -1 & 0 & 0 & -1 & : & -30 \\ 0 & 0 & 0 & 1 & -1 & : & 10 \end{bmatrix}$$

$$R_{23} \text{ and then } R_{34} \sim \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & : & 20 \\ 0 & 1 & 1 & 0 & 0 & : & 20 \\ 0 & -1 & 0 & 0 & -1 & ; & -30 \\ 0 & 0 & 1 & -1 & 0 & : & -20 \\ 0 & 0 & 0 & 1 & -1 & : & 10 \end{bmatrix}$$

$$R_3 + R_2 \sim \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & : & 20 \\ 0 & 1 & 1 & 0 & 0 & : & 20 \\ 0 & 0 & 1 & 0 & -1 & : & -10 \\ 0 & 0 & 1 & -1 & 0 & : & -20 \\ 0 & 0 & 0 & 1 & -1 & : & 10 \end{bmatrix}$$

$$R_4 - R_3 \sim \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & : & 20 \\ 0 & 1 & 1 & 0 & 0 & : & 20 \\ 0 & 0 & 1 & 0 & -1 & : & -10 \\ 0 & 0 & 0 & -1 & 1 & : & -10 \\ 0 & 0 & 0 & 1 & -1 & : & 10 \end{bmatrix}$$

$$-1\,R_4 \sim \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & : & 20 \\ 0 & 1 & 1 & 0 & 0 & : & 20 \\ 0 & 0 & 1 & 0 & -1 & : & -10 \\ 0 & 0 & 0 & 1 & -1 & : & 10 \\ 0 & 0 & 0 & +1 & -1 & : & 10 \end{bmatrix}$$

$$R_5 - R_4 \sim \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & : & 20 \\ 0 & 1 & 1 & 0 & 0 & : & 20 \\ 0 & 0 & 1 & 0 & -1 & : & -10 \\ 0 & 0 & 0 & 1 & -1 & : & 10 \\ 0 & 0 & 0 & 0 & 0 & : & 0 \end{bmatrix}$$

Resulting equations are

$I_1 + I_2 = 20$, $I_2 + I_3 = 20$, $I_3 - I_5 = -10$, $I_4 - I_5 = 10$

Let $I_5 = t$, $I_4 = 10 + t$, $I_3 = -10 + t$

$I_2 = 20 - I_3 = 20 + 10 - t = 30 - t$, $I_1 = 20 - I_2$, $I_1 = 20 - 30 + t = -10 + t$

Here $r[A : B] = r[A]$ But $r < 5$.

∴ System has infinitely many solutions which shows that as we take different values for I_5, we will get different values for I_1, I_2, I_3 and I_4.

Ex. 4 : Determine the currents in the network given in the Fig. 8.6 below.

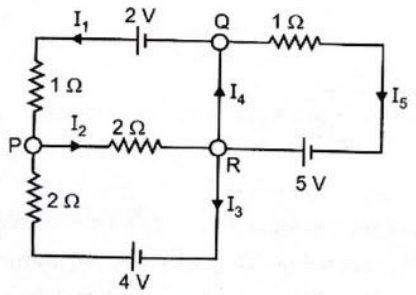


Fig. 8.6

Sol. : Applying KCL at the nodes P, Q, R

$$I_1 + I_3 = I_2 \quad \ldots (1)$$

$$I_1 + I_5 = I_4 \quad \ldots (2)$$

$$I_2 + I_5 = I_3 + I_4 \quad \ldots (3)$$

Applying KVL to three loops

$$I_1 + 2I_2 = 2 \quad \ldots (4)$$

$$2I_3 + 2I_2 = 4 \quad \ldots (5)$$

$$I_5 = 5 \quad \ldots (6)$$

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$I_5 = 5$ is known. We require four equations to obtain four unknowns I_1, I_2, I_3, I_4. We omit equation (3) and proceed to solve the system of equations (1), (2), (4) and (5) we write down the system in matrix form

$$\begin{bmatrix} 1 & -1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 2 \\ 2 \end{bmatrix}$$

$$AX = B \quad \text{where, } X = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix}, \; B = \begin{bmatrix} 0 \\ -5 \\ 2 \\ 2 \end{bmatrix}$$

Consider augmented matrix

$$[A : B] = \begin{bmatrix} 1 & -1 & 1 & 0 & : & 0 \\ 1 & 0 & 0 & -1 & : & -5 \\ 1 & 2 & 0 & 0 & : & 2 \\ 0 & 1 & 1 & 0 & : & 2 \end{bmatrix} \sim \begin{matrix} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{matrix} \begin{bmatrix} 1 & -1 & 1 & 0 & : & 0 \\ 0 & 1 & -1 & -1 & : & -5 \\ 0 & 0 & 2 & 3 & : & 17 \\ 0 & 0 & 2 & 1 & : & 7 \end{bmatrix} \quad R_4 - R_3 \sim \begin{bmatrix} 1 & -1 & 1 & 0 & : & 0 \\ 0 & 1 & -1 & -1 & : & -5 \\ 0 & 0 & 2 & 3 & : & 17 \\ 0 & 0 & 0 & -2 & : & -10 \end{bmatrix}$$

Resulting system is $I_1 - I_2 + I_3 = 0$, $I_2 - I_3 - I_4 = 5$, $2I_3 + 3I_4 = 17$, $-2I_4 = -10$

$I_4 = 5$, $I_3 = 1$, $I_2 = 1$, $I_1 = 0$, $I_5 = 5$

It satisfies equation (3) also.

Ex. 5 : *For a traffic flow problem, ensuring that at each junction total inflow = total out flow of traffic, find the values of unknowns, corresponding to flow pattern shown below.*

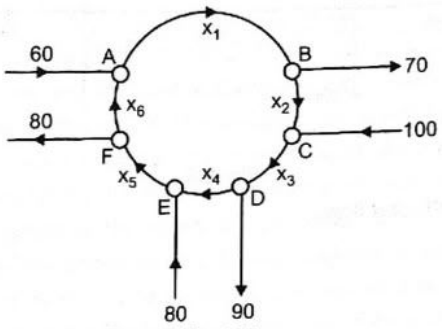


Fig. 8.7

Sol. :

At the junction A,	$x_6 + 60 = x_1$	or	$x_1 - x_6 = 60$	... (1)
At the junction B,	$x_1 - 70 = x_2$	or	$x_1 - x_2 = 70$	... (2)
At the junction C,	$x_2 + 100 = x_3$	or	$x_3 - x_2 = 100$	... (3)
At the junction D,	$x_3 - 90 = x_4$	or	$x_4 - x_3 = -90$	... (4)
At the junction E,	$x_4 + 80 = x_5$	or	$x_5 - x_4 = 80$	... (5)
At the junction F,	$x_5 - 80 = x_6$	or	$x_6 - x_5 = -80$	... (6)

Matrix form of these equation is

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 60 \\ 70 \\ 100 \\ 90 \\ 80 \\ 80 \end{bmatrix}$$

$$[A \mid B] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 & : & 60 \\ 1 & -1 & 0 & 0 & 0 & 0 & : & 70 \\ 0 & -1 & 1 & 0 & 0 & 0 & : & 100 \\ 0 & 0 & 1 & -1 & 0 & 0 & : & 90 \\ 0 & 0 & 0 & -1 & 1 & 0 & : & 80 \\ 0 & 0 & 0 & 0 & 1 & -1 & : & 80 \end{bmatrix} \begin{matrix} \\ -R_2 \\ \\ \\ -R_4 \\ \end{matrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 & : & 60 \\ -1 & 1 & 0 & 0 & 0 & 0 & : & -70 \\ 0 & -1 & 1 & 0 & 0 & 0 & : & 100 \\ 0 & 0 & -1 & 1 & 0 & 0 & : & -90 \\ 0 & 0 & 0 & -1 & 1 & 0 & : & 80 \\ 0 & 0 & 0 & 0 & -1 & 1 & : & -80 \end{bmatrix}$$

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$$\begin{matrix} R_2 + R_1 \\ R_3 + R_2 \\ R_4 + R_3 \\ R_4 + R_5 \\ R_6 + R_5 \end{matrix} \sim \left[\begin{array}{cccccc:c} 1 & 0 & 0 & 0 & 0 & -1 & 60 \\ 0 & 1 & 0 & 0 & 0 & -1 & -10 \\ 0 & 0 & 1 & 0 & 0 & 0 & 90 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 80 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

Resulting equation are

$$\begin{aligned} x_1 - x_6 &= 60 \\ x_2 - x_6 &= -10 \\ x_3 &= 90 \\ x_4 &= 0 \end{aligned} \qquad \begin{aligned} x_5 &= 80 \\ x_6 &= 0 \\ x_1 &= 60 \\ x_2 &= -10 \end{aligned}$$

Ex. 6 : *Solve the following traffic problem.*

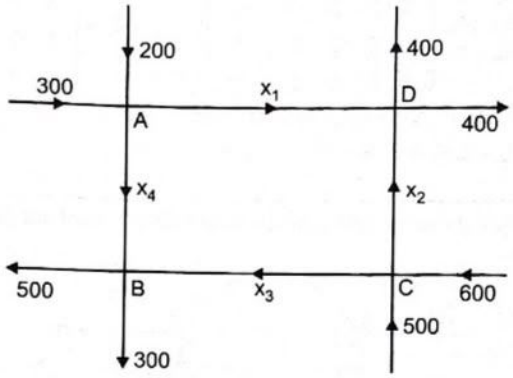


Fig. 8.8

Sol. : Traffic inflow at each junction = Traffic out flow

At junction A, $200 + 300 = x_1 + x_4$

At junction B, $x_3 + x_4 = 500 + 300$

At junction C, $500 + 600 = x_2 + x_3$

At junction D, $x_1 + x_2 = 400 + 400$

The system of equation is

$$\begin{aligned} x_1 + 0x_2 + 0x_3 + x_4 &= 500 \\ x_1 + x_2 + 0x_3 + 0x_4 &= 800 \\ 0x_1 + x_2 + x_3 + 0x_4 &= 1100 \\ 0x_1 + 0x_1 + x_3 + x_4 &= 800 \end{aligned}$$

$$[A : B] = \left[\begin{array}{cccc:c} 1 & 0 & 0 & 1 & 500 \\ 1 & 1 & 0 & 0 & 800 \\ 0 & 1 & 1 & 0 & 1100 \\ 0 & 0 & 1 & 1 & 800 \end{array}\right] \begin{matrix} R_2 - R_1 \sim \\ R_3 - R_2 \sim \\ R_4 - R_3 \sim \end{matrix} \left[\begin{array}{cccc:c} 1 & 0 & 0 & 1 & 500 \\ 0 & 1 & 0 & -1 & 300 \\ 0 & 0 & 1 & 1 & 800 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

We get

$$\begin{aligned} x_1 + x_4 &= 500 \\ x_2 - x_4 &= 300 \\ x_3 + x_4 &= 800 \end{aligned}$$

Let $x_4 = t$ $x_3 = 800 - t$ $x_2 = 300 + t$ $x_1 = 500 - t$

where t can take any value.

If t = 100. $x_1 = 400$, $x_2 = 400$, $x_3 = 700$, $x_4 = 100$

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EXERCISES 8.11

1. Find the currents I_1, I_2 and I_3 in the circuit shown in the figure.

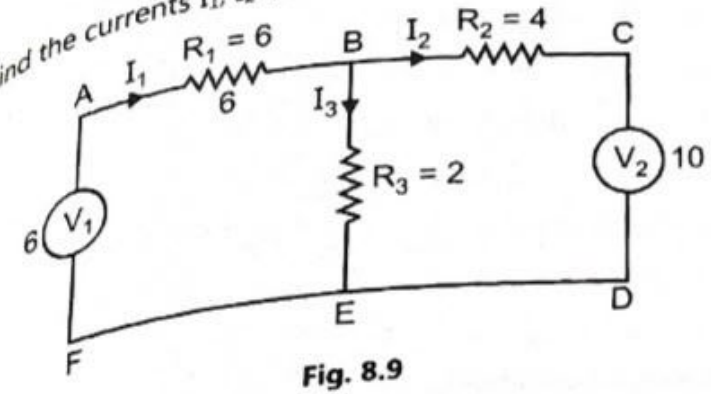


Fig. 8.9

Ans. $I_1 = \frac{8}{5}$, $I_2 = \frac{17}{5}$, $I_3 = -\frac{9}{5}$

3.

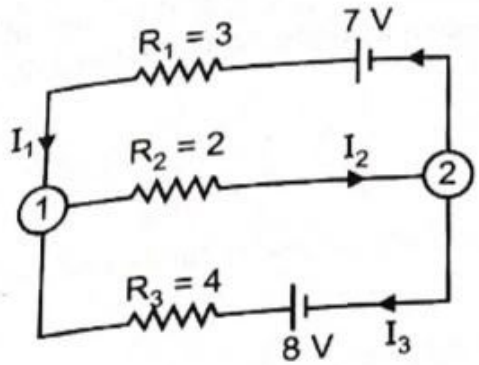


Fig. 8.10

Ans. $I_1 = 1, I_2 = 2, I_3 = 1$

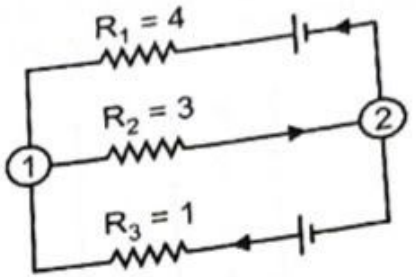


Fig. 8.11

Ans. $I_1 = 0, I_2 = 1, I_3 = 1$

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